Vector 3D: Formula Sheet

Vector basics

Unit vector:

$$\hat{a} = \frac{\vec{a}}{|\vec{a}|}$$

Magnitude:

$$|x\hat{i} + y\hat{j} + z\hat{k}| = \sqrt{x^2 + y^2 + z^2}$$

Given length k: $k\hat{a}$

Collinear:

$$\vec{a} = \lambda\vec{b} \Leftrightarrow \frac{a_1}{b_1} = \frac{a_2}{b_2} = \frac{a_3}{b_3}$$

Triangle inequality:

$$\big||\vec{a}| - |\vec{b}|\big| \le |\vec{a} + \vec{b}| \le |\vec{a}| + |\vec{b}|$$

Section:

$$\vec{r} = \frac{m\vec{b} + n\vec{a}}{m + n}$$

Centroid:

$$\frac{\vec{a} + \vec{b} + \vec{c}}{3}$$

From x-axis: $\sqrt{y^2 + z^2}$

Collinear points:

$$x\vec{a} + y\vec{b} + z\vec{c} = \vec{0},\ x + y + z = 0$$

Dot and cross product

Dot:

$$\vec{a} \cdot \vec{b} = |\vec{a}||\vec{b}| \cos\theta = a_1b_1 + a_2b_2 + a_3b_3$$

Projection:

$$\frac{\vec{a} \cdot \vec{b}}{|\vec{b}|}$$

Component along $\vec{b}$:

$$\frac{\vec{a} \cdot \vec{b}}{|\vec{b}|^2}\vec{b}$$

Expansion:

$$|\vec{a} \pm \vec{b}|^2 = |\vec{a}|^2 + |\vec{b}|^2 \pm 2\vec{a} \cdot \vec{b}$$

Cross:

$$\vec{a} \times \vec{b} = |\vec{a}||\vec{b}| \sin\theta\, \hat{n}$$

Determinant:

$$\vec{a} \times \vec{b} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix}$$

Lagrange:

$$|\vec{a} \times \vec{b}|^2 + (\vec{a} \cdot \vec{b})^2 = |\vec{a}|^2|\vec{b}|^2$$

Unit normal:

$$\pm\frac{\vec{a} \times \vec{b}}{|\vec{a} \times \vec{b}|}$$

Triangle area: $\frac{1}{2}|\vec{AB} \times \vec{AC}|$

Quadrilateral area: $\frac{1}{2}|\vec{d}_1 \times \vec{d}_2|$

Angle bisectors:

$$\lambda(\hat{a} + \hat{b}),\ \lambda(\hat{a} - \hat{b})$$

Direction cosines and lines

DC identity: $l^2 + m^2 + n^2 = 1$

Sines:

$$\sin^2\alpha + \sin^2\beta + \sin^2\gamma = 2$$

DC from DR:

$$\frac{\pm(a, b, c)}{\sqrt{a^2 + b^2 + c^2}}$$

Angle:

$$\cos\theta = \frac{|a_1a_2 + b_1b_2 + c_1c_2|}{\sqrt{\sum a_1^2}\sqrt{\sum a_2^2}}$$

Vector line: $\vec{r} = \vec{a} + t\vec{b},\ \ \vec{r} \times \vec{b} = \vec{a} \times \vec{b}$

Symmetric:

$$\frac{x - x_1}{a} = \frac{y - y_1}{b} = \frac{z - z_1}{c}$$

Two-point:

$$\frac{x - x_1}{x_2 - x_1} = \frac{y - y_1}{y_2 - y_1} = \frac{z - z_1}{z_2 - z_1}$$

Point-line distance:

$$\frac{|(\vec{p} - \vec{a}) \times \vec{b}|}{|\vec{b}|}$$

SD skew:

$$\frac{|(\vec{a}_2 - \vec{a}_1) \cdot (\vec{b}_1 \times \vec{b}_2)|}{|\vec{b}_1 \times \vec{b}_2|}$$

SD parallel:

$$\frac{|(\vec{a}_2 - \vec{a}_1) \times \vec{b}|}{|\vec{b}|}$$

Intersect:

$$[\vec{a}_2 - \vec{a}_1 \ \ \vec{b}_1 \ \ \vec{b}_2] = 0$$

Plane

Normal form: $\vec{r}\cdot\hat{n}=d$

Point-normal: $a(x-x_1)+b(y-y_1)+c(z-z_1)=0$

Intercept:

$$\frac{x}{a}+\frac{y}{b}+\frac{z}{c}=1$$

Parametric:

$$\vec{r}=\vec{a}+\lambda\vec{b}+\mu\vec{c}$$

Three points:

$$[\vec{r}-\vec{a}\ \ \vec{b}-\vec{a}\ \ \vec{c}-\vec{a}]=0$$

Planes angle:

$$\cos\theta=\frac{|\vec{n}_1\cdot\vec{n}_2|}{|\vec{n}_1||\vec{n}_2|}$$

Line-plane angle:

$$\sin\theta=\frac{|\vec{b}\cdot\vec{n}|}{|\vec{b}||\vec{n}|}$$

Line in plane:

$$\vec{b}\cdot\vec{n}=0,\ \vec{a}\cdot\vec{n}=d$$

Distance, foot, image

Point-plane:

$$\frac{|ax_1+by_1+cz_1+d|}{\sqrt{a^2+b^2+c^2}}$$

Parallel planes:

$$\frac{|d_1-d_2|}{\sqrt{a^2+b^2+c^2}}$$

Foot:

$$\frac{x-x_1}{a}=\frac{y-y_1}{b}=\frac{z-z_1}{c}=-\frac{ax_1+by_1+cz_1+d}{a^2+b^2+c^2}$$

Image:

$$\frac{x-x_1}{a}=\frac{y-y_1}{b}=\frac{z-z_1}{c}=-\frac{2(ax_1+by_1+cz_1+d)}{a^2+b^2+c^2}$$

Same side: $P_1(x_1)\,P_1(x_2)>0$

Bisector planes:

$$\frac{a_1x+b_1y+c_1z+d_1}{\sqrt{\sum a_1^2}}=\pm\frac{a_2x+b_2y+c_2z+d_2}{\sqrt{\sum a_2^2}}$$

Origin bisector obtuse:

$$d_1,d_2>0,\ a_1a_2+b_1b_2+c_1c_2>0$$

Family, intersection, images

Family: $P_1+\lambda P_2=0$

Line of intersection: $\vec{b}=\vec{n}_1\times\vec{n}_2$

Line and point:

$$\vec{n}=(\vec{p}-\vec{a})\times\vec{b}$$

Two lines:

$$[\vec{r}-\vec{a}_1\ \ \vec{b}_1\ \ \vec{b}_2]=0$$

Parallel lines:

$$\vec{n}=(\vec{a}_2-\vec{a}_1)\times\vec{b}$$

Greatest slope:

$$(\vec{n}_H\times\vec{n}_G)\times\vec{n}_G$$

Triple products

STP:

$$[\vec{a}\ \vec{b}\ \vec{c}]=\vec{a}\cdot(\vec{b}\times\vec{c})$$

Sum identity:

$$[\vec{a}+\vec{b}\ \ \vec{b}+\vec{c}\ \ \vec{c}+\vec{a}]=2[\vec{a}\ \vec{b}\ \vec{c}]$$

Cross identity:

$$[\vec{a}\times\vec{b}\ \ \vec{b}\times\vec{c}\ \ \vec{c}\times\vec{a}]=[\vec{a}\ \vec{b}\ \vec{c}]^2$$

Gram:

$$[\vec{a}\ \vec{b}\ \vec{c}]^2=\begin{vmatrix}\vec{a}\cdot\vec{a} & \vec{a}\cdot\vec{b} & \vec{a}\cdot\vec{c}\\ \vec{b}\cdot\vec{a} & \vec{b}\cdot\vec{b} & \vec{b}\cdot\vec{c}\\ \vec{c}\cdot\vec{a} & \vec{c}\cdot\vec{b} & \vec{c}\cdot\vec{c}\end{vmatrix}$$

Parallelepiped: $V=|[\vec{a}\ \vec{b}\ \vec{c}]|$

Tetrahedron:

$$V=\tfrac{1}{6}|[\vec{AB}\ \vec{AC}\ \vec{AD}]|$$

Regular tetrahedron:

$$V=\frac{a^3}{6\sqrt{2}},\ h=a\sqrt{\tfrac{2}{3}}$$

Coplanar: $[\vec{a}\ \vec{b}\ \vec{c}]=0$

VTP:

$$\vec{a}\times(\vec{b}\times\vec{c})=(\vec{a}\cdot\vec{c})\vec{b}-(\vec{a}\cdot\vec{b})\vec{c}$$

Dot of crosses:

$$(\vec{a}\times\vec{b})\cdot(\vec{c}\times\vec{d})=(\vec{a}\cdot\vec{c})(\vec{b}\cdot\vec{d})-(\vec{a}\cdot\vec{d})(\vec{b}\cdot\vec{c})$$

Cross of crosses:

$$(\vec{a}\times\vec{b})\times(\vec{c}\times\vec{d})=[\vec{a}\ \vec{b}\ \vec{d}]\vec{c}-[\vec{a}\ \vec{b}\ \vec{c}]\vec{d}$$

Linear combination and equations

Coefficient:

$$x=\frac{[\vec{r}\ \vec{b}\ \vec{c}]}{[\vec{a}\ \vec{b}\ \vec{c}]}$$

Reciprocal:

$$\vec{a}'=\frac{\vec{b}\times\vec{c}}{[\vec{a}\ \vec{b}\ \vec{c}]},\ [\vec{a}'\ \vec{b}'\ \vec{c}'][\vec{a}\ \vec{b}\ \vec{c}]=1$$

$\vec{r}\times\vec{a}=\vec{b}$:

$$\vec{r}=\frac{\vec{a}\times\vec{b}}{|\vec{a}|^2}+\lambda\vec{a}$$

$\vec{r}\cdot\vec{a}=p,\ \vec{r}\times\vec{b}=\vec{c}$:

$$\vec{r}=\frac{\vec{a}\times\vec{c}+p\vec{b}}{\vec{a}\cdot\vec{b}}$$

$\vec{x}+\vec{a}\times\vec{x}=\vec{b}$:

$$\vec{x}=\frac{\vec{b}+(\vec{a}\cdot\vec{b})\vec{a}-\vec{a}\times\vec{b}}{1+|\vec{a}|^2}$$